Anatomy as a bridge

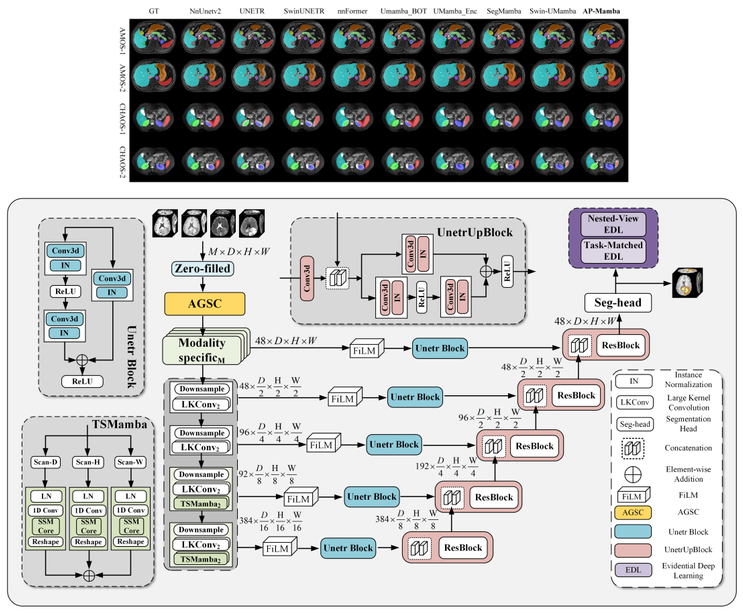
Hekai Wang / Medical imaging & machine learning

How can anatomical knowledge help us understand images, recover missing information, and judge the evidence behind a report?

01 UNDERSTAND

Anatomy-aware understanding

How can structural knowledge guide segmentation across organs and available MRI modalities?



AP-Mamba: multi-organ predictions / ReLiSS: architecture

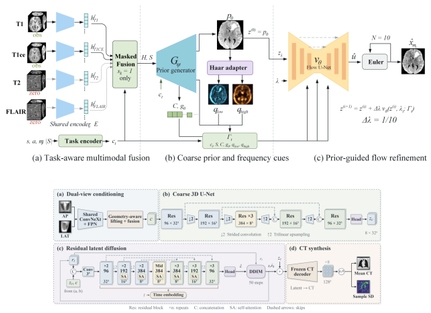
AP-Mamba + ReLiSS

Anatomical priors and state space models for segmentation; modality-adaptive conditioning when MRI inputs vary.

02 RECOVER

Learning with incomplete inputs

What can be recovered from missing MRI sequences or only two X-ray views?



PriorFM: prior-to-target transport / UPath-X2CT: geometry + diffusion

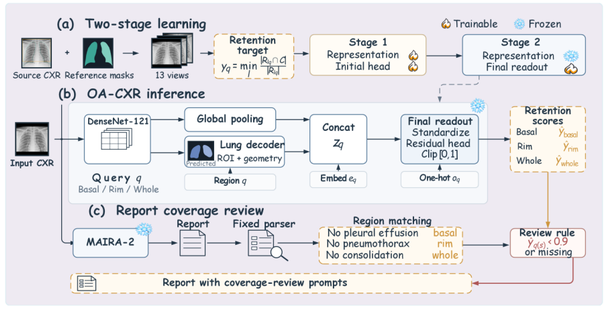
PriorFM + UPath-X2CT

Prior-guided flow matching for MRI synthesis; geometric lifting and residual latent diffusion for biplanar reconstruction.

03 REVIEW

Image-language consistency

Does a presented radiograph retain the anatomical evidence needed for a report statement?



OA-CXR: anatomical retention and selective report review

OA-CXR

Estimate source-relative anatomical retention, then map report statements to regional queries to prioritize review.

THE CONNECTING QUESTION

How much anatomical information is available, and how should a model use it?

Paul C. Lauterbur Research Center for Biomedical Imaging
Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences
University of Chinese Academy of Sciences | Shenzhen, China

hk.wang2@siat.ac.cn
github.com/hekaiwang

Research overview; individual studies remain distinct.